

		(9.17×10^{-30} kg for more massive star)		
			Total	[12]
4	(a)	e.g. amplitude is not constant or wave is damped <i>do not allow 'displacement constant'</i> should be $(-)\cos$, (not $\sin$)	B1	
			B1	[2]
	(b)	$T = 0.60$ s $\omega = 2\pi/T = 10.5 \text{ rad s}^{-1}$ (<i>allow $10.4 \rightarrow 10.6$</i>)	C1 A1	[2]
	(c)	same period displacement always less amplitude reducing appropriately <i>for 2nd and 3rd marks, ignore the first quarter period</i>	B1 M1 A1	[3]
			Total	[7]